15. Seeds and grains — from seed to bread, and saving seed for next year

Oracle OS guide. Manuals in the library: FAO (community seeds, seed production), "Home Vegetable Gardening", "The Vegetable Garden", "War Gardens", "Ten Acres Enough", Practical Action (irrigation, pumps), Gutenberg agriculture (illustrated, in Encyclopedias).

1. Which seeds to keep

- **Open-pollinated / heirloom** (not "F1" hybrids): only these breed true. Label each batch with name, year and place.
- Grains: wheat, rye (tolerates poor soil and cold), barley, oats, maize. Legumes: beans, peas, broad beans, chickpeas, lentils (fix nitrogen). Roots: **potatoes** (most calories per m²), sweet potato (warm climates), Jerusalem artichoke. Squash, cabbages, onion, garlic, carrot, beetroot, tomato, pepper. Herbs: parsley, coriander, thyme.
- Store in dry, cool, dark jars; typical life 2–5 years (onion 1–2, tomato and beans 4–6). Germination test: 10 seeds on damp paper; how many sprout in 7–10 days = germination %.

2. Soil fertility without chemical fertiliser

- **Compost**: layers of green (kitchen scraps, grass) and brown (dry leaves, straw) 1:2, damp like a wrung sponge, turn every 2–3 weeks; ready in 3–6 months.
- Well-rotted **manure** (chicken, cow, horse); never fresh on leaves eaten raw.
- **4-year rotation**: legumes → leafy (brassicas) → roots → potatoes/grain; prevents disease and soil exhaustion.
- **Mulch** (straw, leaves): keeps moisture, smothers weeds.
- **Green manure**: sow clover/field beans/lupins and dig in before flowering.
- Wood ash (potassium, small amounts), crushed eggshells (calcium).

3. Calendar (northern hemisphere, temperate)

- **Winter** (Dec–Feb): plan, repair tools, sow under cover (onions), prune fruit trees.
- **Spring** (Mar–May): potatoes (Mar–Apr), peas, broad beans, carrots, lettuce, cabbages; after frosts: beans, maize, squash, tomatoes.
- **Summer** (Jun–Aug): water early morning, harvest, save seed, sow winter brassicas.
- **Autumn** (Sep–Nov): **winter wheat and rye** (Oct–Nov), garlic, broad beans; harvest squash and potatoes; store.
- Southern Portugal ~3–4 weeks earlier; Scotland 2–4 weeks later.

4. From grain to bread

1. **Harvest** when the grain is hard and the straw yellow (barely dents with a fingernail). Cut with sickle/scythe, tie in sheaves, dry 1–2 weeks in stooks.
2. **Thresh**: beat the sheaves with a flail on a clean cloth or floor.

3. **Winnow**: toss the grain in a light breeze; chaff blows away, grain falls.
4. **Store** in sealed containers against rats and damp (drums, lidded clay pots); dry grain keeps for years.
5. **Grind**: hand quern (two flat stones, the upper one turns), water/wind mill, or mortar. Sieve for finer flour.
6. **Sourdough starter**: flour + water 1:1, stir and "feed" daily for 5–7 days until bubbly and sour-smelling; keep a little from each dough for the next.
7. **Bread**: 1 kg flour + 600–700 ml water + 20 g salt + 200 g starter; knead 10 min, rise 4–12 h, shape, bake in a wood oven (~220–250 °C) 40–50 min. See "A Treatise on Bread" in the library.
8. **Maize**: nixtamalise (cook with lime or wood ash) to prevent pellagra if it is a staple.

5. Saving seed from your own plants

- Pick the **best plants** (healthy, productive, survivors) and let them ripen fully.
- **Dry seeds** (beans, peas, grains, lettuce, onion): dry on the plant, thresh, dry 1–2 more weeks in shade.
- **Wet seeds** (tomato, squash, cucumber): scoop out; tomato ferments 2–3 days in water, then rinse and dry.
- Keep varieties of the same species that cross (maize, squash, brassicas) apart by distance or timing.
- Potatoes and garlic: keep the best tubers/bulbs for planting.

6. Watering

Drip irrigation from punctured bottles, buried clay pots (ollas), rainwater harvesting, water at dusk or dawn. See Practical Action (micro-irrigation, treadle pump).